

ENGINEERING

Ultrasoft hydrogel immune millirobot with multimodal locomotion

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Advancements in cellular immunotherapy demanded efficient immune cell delivery. To meet this need, we introduced hydrogel-based immune millirobots designed for high immune cell loading and precise tumor targeting. These ultrasoft robots, embedded with magnetic nanoparticles, exhibited adaptable locomotion: walking, rolling, climbing, and undulating, enabling navigation through complex biological environments and alignment with varied tumor morphologies. They responded to magnetic fields and ionic or pH changes, facilitating propulsion, grasping, and localized delivery. In vitro, the millirobots eradicated three-dimensional tumor models in four days; in vivo, they notably reduced tumor growth in HepG2-luc tumor-bearing nude mice within 15 days. Bioluminescence imaging confirmed enhanced natural killer cell activity at tumor sites. The robots demonstrated excellent biocompatibility and biodegradability and caused no adverse effects postimplantation. This work showcased a responsive, soft robotic system with potential for advancing immune cell delivery and exploring tumor-immune dynamics in cancer therapy.

INTRODUCTION

In regenerative biology and immunology, cell therapy has undergone rapid advancements (1). Cell therapy, encompassing stem cell therapy and immunotherapy, has shown promise in clinical research (2–4). Autologous stem cell therapy, for example, has been widely applied in treating neurodegenerative (5) and retinal degenerative diseases (6). Immunotherapy, especially chimeric antigen receptor T cell (CAR T cell) therapy (7), has gained traction not only in hematological tumors like lymphoma, myeloma, and leukemia but also in solid tumors such as those in the digestive tract. In immunotherapy, the abundance and proximity of therapeutic cells are crucial for effective treatment (8, 9). However, the limited migration of therapeutic cells to tumor sites remains a notable challenge, hindering treatment efficacy (10, 11). Therefore, an efficient and safe cell carrier is needed to achieve the local accumulation and precise delivery of the therapeutic cells.

Untethered soft miniature robots (USMRs) have garnered increasing attention in various fields that demand not only precise sensing (12), dynamic actuation (13), and versatile manipulation (14) but also effective solutions for drug delivery (15, 16). USMRs, with their modular functional design, are capable of performing multiple tasks, including tissue scaffold construction (17), biosensing (18, 19), minimally invasive surgery (20–22), targeted cell delivery (23–25), and drug delivery (21, 26). Despite their potential, traditional USMRs encounter mobility constraints (27) and are typically limited by a single response mode (28), making it difficult to balance robot size (25, 29), delivery efficiency (30), and control flexibility (31)

particularly in complex bio-microenvironments where obstacles and texture changes pose notable challenges (23, 32).

To overcome these limitations, the softness of USMRs plays a crucial role in enhancing their adaptability and functionality in biological environments (33, 34). A softer structure enables better compliance with the intricate and irregular surfaces of tissues, reducing mechanical damage and improving biocompatibility (35–37). This flexibility allows USMRs to navigate through narrow and dynamically changing biological pathways, ensuring precise and efficient cell delivery (38, 39). Additionally, increased softness enhances the robots' ability to conform to lesion sites, maximizing the contact area between therapeutic cells and target tissues, which is essential for effective immunotherapy (40, 41). By mimicking the natural flexibility and adaptability of biological organisms, ultrasoft USMRs can improve their maneuverability, enhance their ability to transport viable therapeutic cells, and optimize treatment outcomes in regenerative medicine and targeted cell therapy (26, 42, 43). Despite progress in magnetic soft locomotion, many existing systems still lack the ability to programmably switch between multiple locomotion modes, adapt effectively to the biomechanical properties of soft tumor tissues, or operate in environments compatible with immune cell viability, which are essential for realizing the full therapeutic potential of USMRs.

To enhance the physical targeting and recruitment of cells in vivo, we designed a hydrogel immune millirobot (HIM) for studying the interaction of tumor-immune cells for investigating targeted delivery for cell therapy. This design attempted to address the challenges of in vivo navigation, which consisted of liquid environments and varied terrains that demanded carriers to be capable of diverse motion and response modes as illustrated in Fig. 1. The HIM was constructed from an ultrasoft electrodeposited hydrogel infused with immune cells, mirroring the functional design of a starfish's mouth region. Strategically positioned within the flat abdominal area of the robot, this structure optimized lesion fitting, increased the contact area between immune cells and target lesions, and facilitated cell migration. To enhance propulsion, magnetic nanoparticles (MNP; FePt) were embedded within the HIM's legs,

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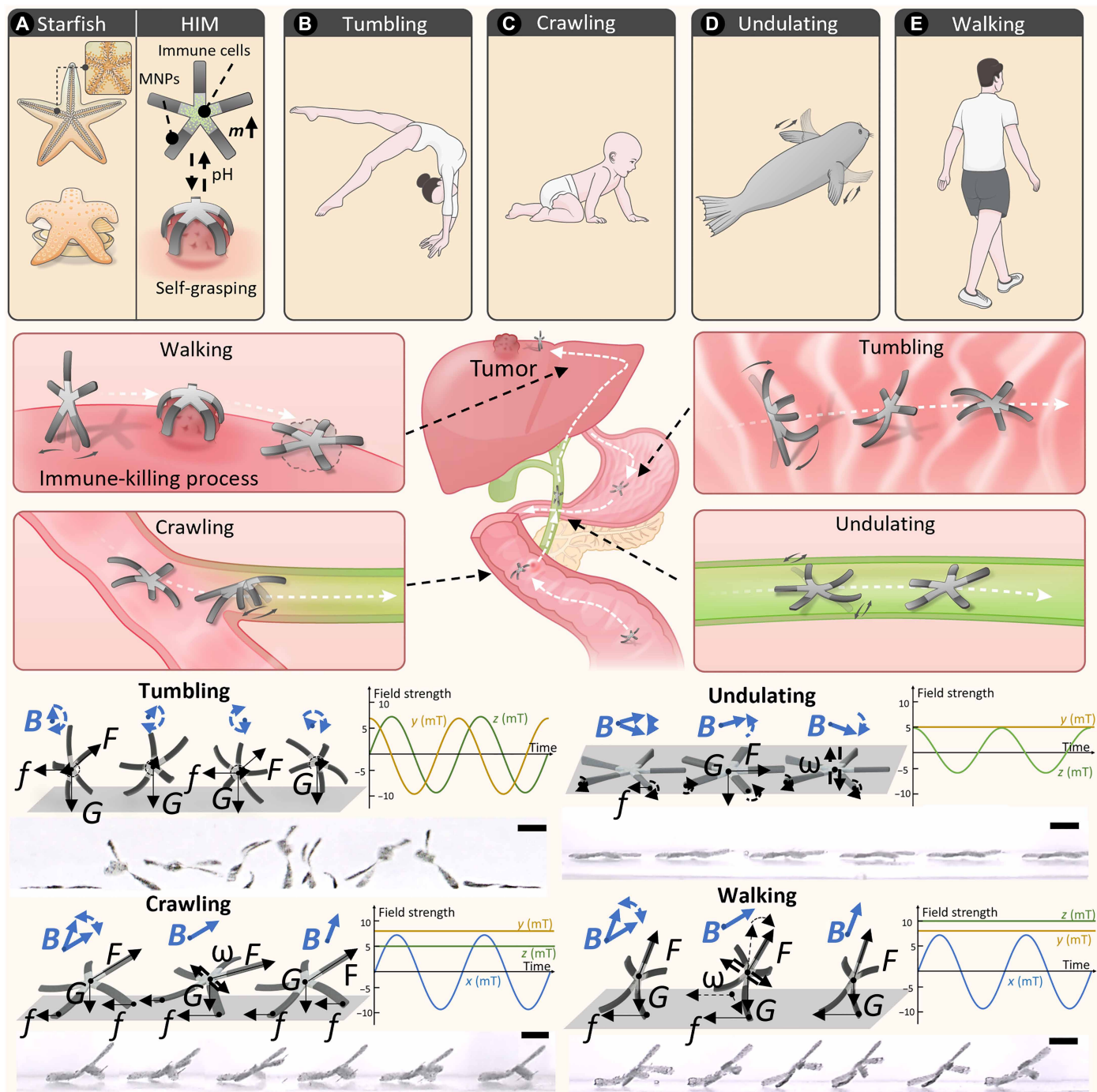


Fig. 1. Modular design, magnetic profile, and multiple locomotion modes of the magnetic HIM. (A) The comparison of real starfish and the HIM with magnetic legs and immune cell core. The starfish-shaped magnetic soft robot's dimensions and magnetization profile m . The profile can be described as a single direction along the long axis of the HIM. (B) The tumbling of HIM is induced by rotating B_{yz} . B represents the combination of oscillating and uniform magnetic fields, ω is the oscillating velocity, G is the gravity, f is the friction of the HIM, and F is the holding force. HIM's magnetic moment tries to align with the rotating field, thus generating a rolling gait. Scale bar, 2 mm. (C) The crawling of HIM is induced by static B_{xz} and rotating B_y . This strategy is similar to walking motion with a minor static field. Therefore, the HIM has four legs in contact with the ground, which provide large friction and actuation force. Scale bar, 2 mm. (D) The undulating of HIM is induced by static B_y and rotating B_z . The static field provides guidance while the rotating field provides forward traction force. Scale bar, 2 mm. (E) The biped walking of HIM is induced by static B_{xz} and rotating B_y . The static field realizes stable upright and body creep and the rotating field alternately lifts and swings two legs of HIM to achieve a rocking forward motion. The process is similar to a human-like biped walk. Scale bar, 2 mm.

mimicking the locomotion mechanics of humans and seals. Magnetized along the long axis of the HIM, the nanoparticles enabled the robot to perform a variety of movements. A comparative analysis, illustrated in Fig. 1A, highlighted the starfish-like design, in which the central section of the HIM harbored immune cells (NK92-MI) while its legs encapsulated MNPs. This modular configuration not only emulated the locomotion and feeding mechanisms of a starfish but also, crucially, separated the movement module from the cell-loading module, helping preserve the activity of the carried cells.

RESULTS

The structure design and magnetic actuation

Through modular electrodeposition techniques, various components were precisely embedded into distinct segments of the HIM (fig. S1). During fabrication, the gradient of current density induced dynamic responses within the hydrogel structure, which were influenced by cations and pH levels, thereby triggering shape morphing of the HIM. Figure S2 further provided insights into the finite element analysis (FEA) of microelectrode current density and the transformation process of the HIM.

The magnetic profile of the HIM, as depicted in Fig. 1A, facilitated four distinct locomotion modes: tumbling, crawling, undulating, and walking, each meticulously demonstrated in Fig. 1 (B to E) and elucidated in movies S1 to S4. The HIM's locomotion modes were efficiently controlled through a time-varying magnetic field (**B**), enabling precise maneuverability in desired directions. Drawing inspiration from the undulating motion of marine mammals such as seals, the HIM incorporated an undulating field within the *yz* plane during locomotion. This design feature allowed the HIM's legs to exert pressure against the ground, effectively generating propulsive force while swiftly adjusting movements, thereby enhancing maneuverability. Furthermore, leveraging insights from the agility of backflips, the HIM demonstrated the capability to execute tumbling motions akin to humans. This is achieved through the generation of rotating fields on either the *xz* or *yz* plane, coupled with the inherent elasticity of the HIM's legs. The resulting spring-like rebound generated the driving force necessary for propulsion, enabling the HIM to traverse obstacles adeptly and navigate challenging terrains with ease. In addition to these advanced locomotion modes, the HIM exhibited a crawling motion reminiscent of walking, with all four legs maintaining contact with the ground. This was facilitated by the generation of a relatively lower B_z magnetic field, enhancing friction and stability while minimizing the risk of slipping.

Multimodal locomotion under magnetic control

In Fig. 2A, the detailed magnetic profile was along the stand-up direction. Its motion was facilitated by the deformation of magnetized antennas in response to the magnetic field. The magnetic moment continuously acted on the HIM's antenna until the antenna's magnetization direction aligned with the external magnetic field. Under the influence of a quasistatic magnetic field, the deformation of the element in the *yz* plane was depicted in Fig. 2B. Therefore, the antenna deformation could be expressed by the fourth-order ordinary differential equation. The main factors affecting the deformation were magnetic field, material, size, and magnetization. The antenna root experienced no deformation, and the antenna free end showed no deflection. Here, we analyzed the deformation of a leg. However, as the deformation was not simple bending when the magnetization

direction was inconsistent with the x_i and z_i directions, FEA was used to analyze the different legs along the other directions. The robot coordinate system (x_R, y_R) and antenna coordinate system (x_i, y_i) were defined in Fig. 2C. In the walking and crawling mode, the HIM adeptly maintained stability through static magnetic forces on the *xz* plane, and, then, the oscillating field was programmed to swing to the left and right directions alternately, maintaining a forward movement, showcasing exceptional attitude controllability (fig. S3). In the tumbling motion, the magnetic moment of the HIM tended to align with the external rotating magnetic field B_{xz} , resulting in a rolling gait (fig. S4). In the undulating motion, the reciprocating magnetic torque along the *z* axis prompted the HIM's magnetic antennas to repeatedly deform and strike the ground (fig. S5), and the deformation recovery of the antenna provided forward traction. Further enhancements in three-dimensional (3D) posture adjustment were demonstrated through the manipulation of uniform fields on the *z* axis and rotation on the *xy* plane (fig. S6 and movie S5). To further showcase the HIM's versatility, we orchestrated movements along trajectories resembling the letters "M," "P," and "I" using walking, undulating, and tumbling modes in Fig. 2D and movie S6, with tumbling emerging as the fastest, achieving a notable speed of 5.65 mm/s (Fig. 2E). The walking, undulating, and crawling speed reached 7.62, 1.73, and 7.11 mm/s, respectively (Fig. 2, F to H). The best working frequency for these four locomotion modes was 3 to 7 Hz. Through these demonstrations, we underscored the HIM's adaptability and precision in traversing complex paths with diverse shapes and orientations, marking an avenue in the field of biomimetic robotics.

Locomotion across various obstacle scenarios

In biological microenvironments, robots encountered uncertain obstacles on their path to reaching target areas, necessitating flexible control methods and high trafficability (44). To showcase the robot's prowess in navigating complex and unstructured biological microenvironments, we presented a comprehensive exploration of its locomotion capabilities across various obstacle scenarios (Fig. 3A and movies S7 to S12). Beginning with its adept negotiation of a 30° slope using walking motion (Fig. 3B), the HIM demonstrated its remarkable ability to handle inclines, mirroring the agility of human and seal mobility. Expanding its repertoire, the HIM exhibited agility by ascending 3-mm-high stairs via tumbling motion (Fig. 3C), facilitated by the integration of MNP in its legs, akin to the dynamic movements observed in HIM. In addition to conquering vertical obstacles, by using undulating and crawling motions, the HIM navigated through 0.8- and 2-mm channels, respectively (Fig. 3D), showcasing versatility inspired by human and seal locomotion mechanisms. The HIM showcased proficiency in navigating confined spaces, deftly sidling through a narrow 2-mm-width slit (Fig. 3E). Leveraging its hydrogel structure and integrated immune cells, reminiscent of starfish anatomy, the HIM adeptly adapted its locomotion strategy to traverse channels of varying widths. Further highlighting its adaptability, the HIM dynamically adjusted its form to pass through a narrowing tunnel with inner diameters ranging from 8 to 2 mm (Fig. 3F), showcasing its supersoftness and ability to respond to changing environmental conditions.

Due to the extremely low stiffness of the HIM, it is difficult to accurately measure its Young's modulus using conventional methods. However, using a customized measurement approach detailed in fig. S7, we determined the Young's modulus to be ~808.5 Pa. An

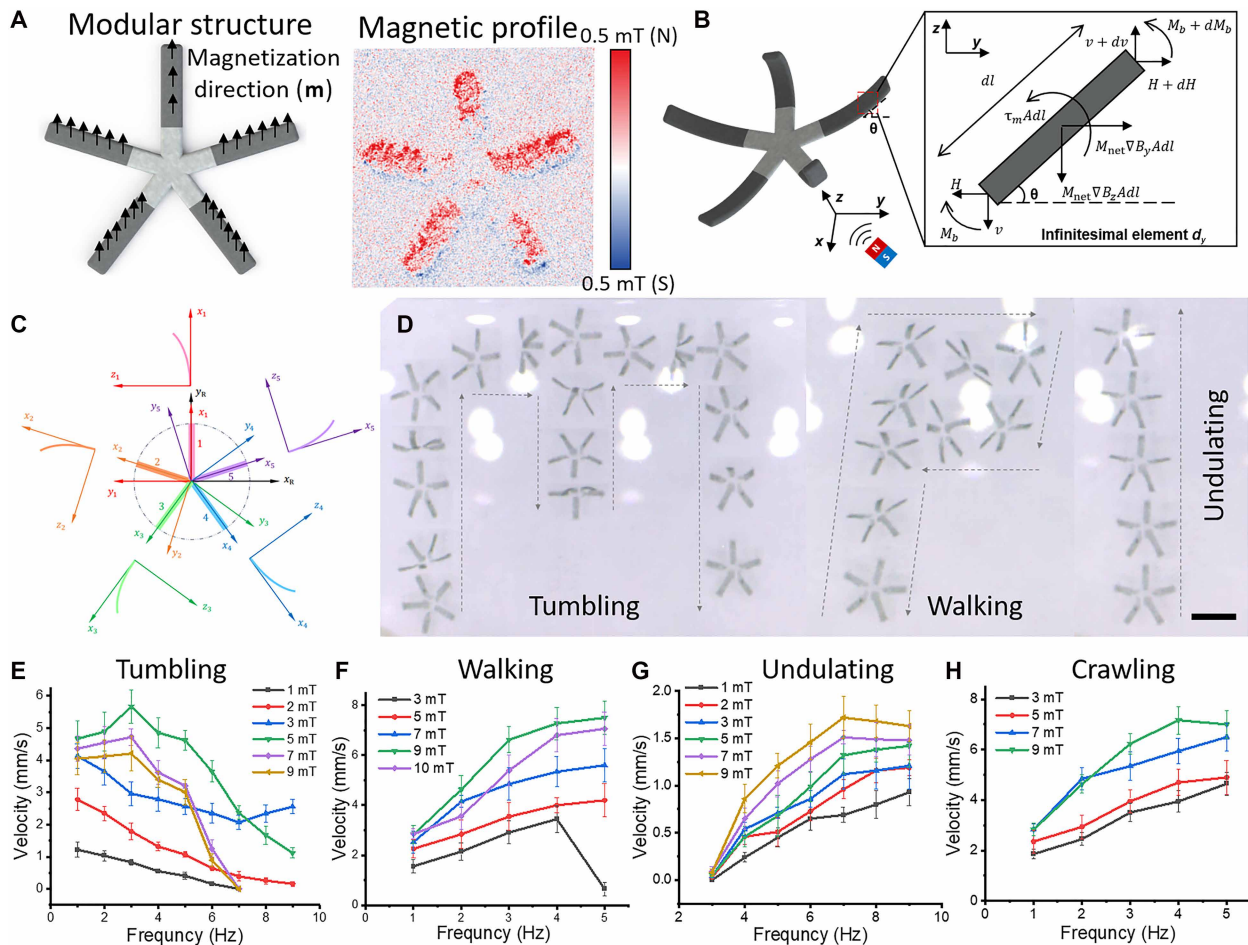


Fig. 2. Elastic-magnetic bending model of the magnetic leg of the HIM. (A) The magnetization direction and magnetic profile of HIM. N, north pole; S, south pole. (B) The bending moment acts on an infinitesimal element of the HIM under steady-state deformation. (C) FEA of five legs in five different directions with 72° interval. (D) Tumbling, walking, and undulating trajectory of HIM with “M,” “P,” and “I” shape, separately. Scale bar, 3 mm. (E) Velocity-frequency profiles for HIM with different magnetic rotating fields (1 to 9 mT) based on tumbling locomotion. Each dot represents the average velocity of three different HIMs $\pm$ SEM. (F) Velocity-frequency profiles for HIM with different magnetic oscillating fields (3 to 10 mT) based on walking locomotion. Each dot represents the average velocity of three different HIMs $\pm$ SEM. (G) Velocity-frequency profiles for HIM with different magnetic oscillating fields (1 to 9 mT) based on undulating locomotion. Each dot represents the average velocity of three different HIMs $\pm$ SEM. (H) Velocity-frequency profiles for HIM with different magnetic rotating fields (3 to 9 mT) based on crawling locomotion. Each dot represents the average velocity of three different HIMs $\pm$ SEM.

analysis of posture and magnetic field strength on stiffness, illustrated in fig. S8, revealed that a larger standing angle and increased magnetic field strength resulted in a straighter body configuration. Due to its supersoft robotic body, the HIM could be injected through a needle with a diameter as small as 1 mm, as depicted in fig. S9 and movie S13. Moreover, the HIM's remarkable damage resistance enabled it to retain locomotion ability even when some of its legs were lost. Last, the HIM successfully maneuvered within a 3-mm-diameter channel (Fig. 3G), demonstrating its adaptability to confined biological spaces that closely mimicked in vivo structures such as gastrointestinal (GI) lumen folds, intestinal crypts, and intertissue clefts in the peritoneal cavity. These features commonly ranged from submillimeter to several millimeters in scale, presented topographical and spatial challenges for targeted delivery systems. The HIM was tested on the surface of fresh stomach tissue in liquid adjusted to a viscosity of 5 mPa·s (representing the upper range of gastric

fluid viscosity observed in the empty stomach). The HIM demonstrated comparable walking-mode locomotion under this elevated viscosity, as shown in fig. S10 and movie S14. The ability of the HIM to traverse these constraints highlighted its biomimetic adaptability and suitability for in vivo navigation through complex, uneven microenvironments. Together with its soft, biocompatible design and immune cell-carrying capability, these results supported the HIM's potential as a robust and versatile robotic platform for biomedical applications, including immune cell-based therapies in anatomically constrained regions.

In vitro immune-killing process

To test the biocompatibility and cell loading capability of the HIM, we first integrated green fluorescent protein (GFP)-enhanced human umbilical vein endothelial cells (HUVECs) into the cell loading module of the HIM (fig. S11). Within 4 days, the cells migrated onto

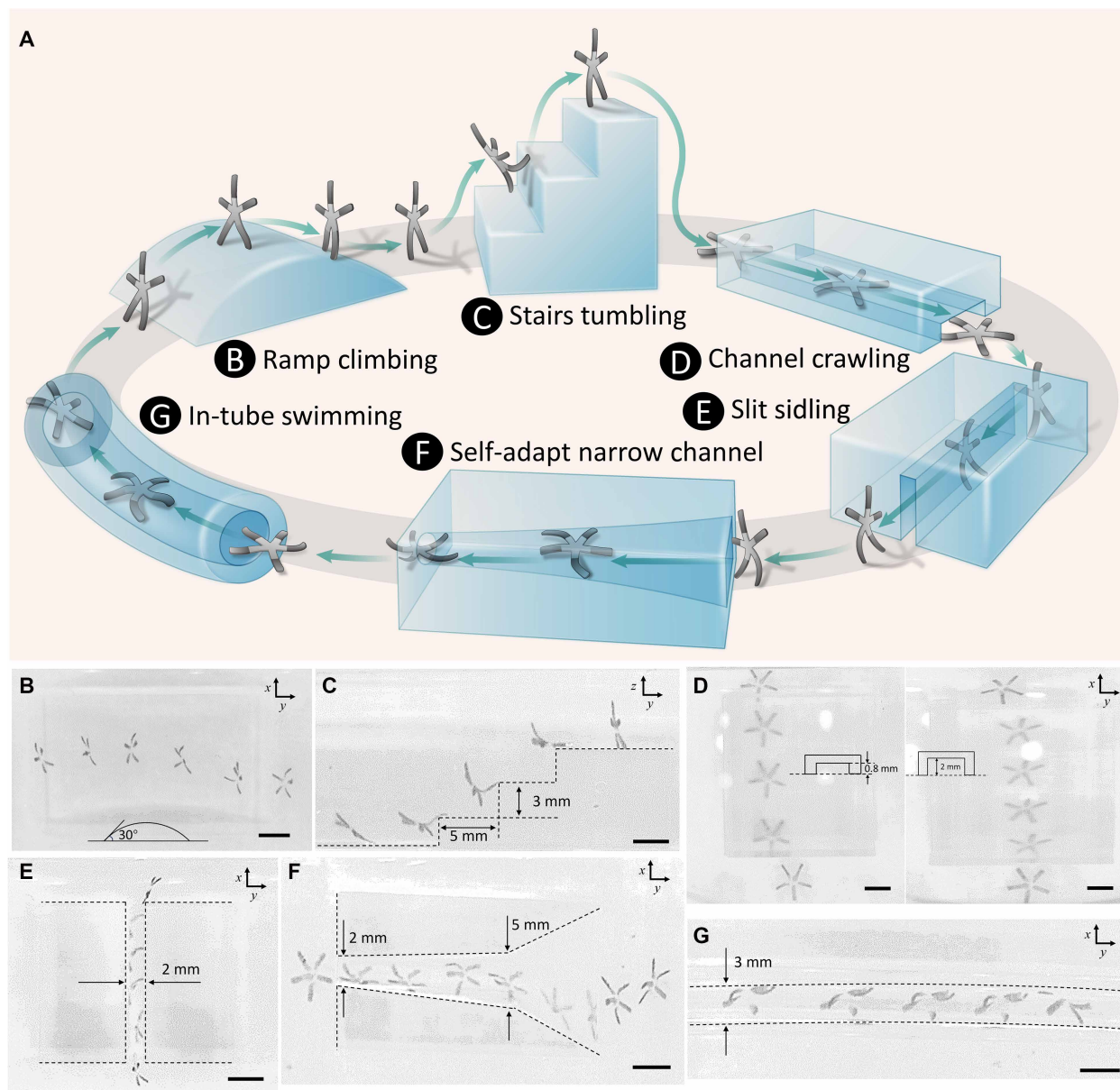


Fig. 3. Multimodal locomotion demonstrated in a complex obstacle environment. (A) Multimodal locomotion modes. (B) The HIM traverses a 30° slope using walking motion. Scale bar, 6 mm. (C) The HIM ascends 3-mm-height stairs using tumbling motion. Scale bar, 6 mm. (D) The HIM maneuvers through 0.8- and 2-mm channels using undulating and crawling motions. Scale bar, 6 mm. (E) The HIM navigates through a 2-mm-width slit with a sidling motion. Scale bar, 6 mm. (F) The HIM dynamically adapts to pass through a narrowing tunnel, with inner diameters ranging from 8 to 2 mm. Scale bar, 5 mm. (G) The HIM moves within a 3-mm-diameter pipe. Scale bar, 5 mm.

the spherical structure integrated within the HIM. Even after the HIM was removed, the remaining HUVECs continued to proliferate and completely covered the spherical structure within 3 days, suggesting the HIM's efficacy and safety as a vehicle for delivering bio-agents. To evaluate retention under dynamic conditions, we subjected the HIM to a simulated GI environment featuring continuous physiological flow (3 ml/min) and aperiodic mechanical perturbation. The HIM maintained robust adhesion to porcine intestinal mucosa for 72 hours under these combined stresses (fig. S12), demonstrating its ability to withstand physiologically relevant shear and motion. In further investigating its potential in cancer therapy, the HIM was loaded with NK92-MI cells in the core for

targeted delivery to the tumor site (schematic in Fig. 4A and movie S15) (45). To ensure that the cell-bearing side of the HIM was oriented correctly toward the target tumor, we used a walking locomotion mode after reaching the tumor for precise positioning (Fig. 4B). Subsequently, the HIM grasped the 3D tumor model in response to pH changes (Fig. 4C). Following the immune-killing process, HIM was exposed to sodium ions (sodium citrate solution) and dissolved (Fig. 4D) which are water soluble and can be absorbed by living organisms. Under confocal microscopy, the HUVECs exhibited a notable decrease in intensity after 4 days of treatment with NK92-cell loaded HIM (Fig. 4E) with a survival rate nearly reaching 0% (Fig. 4F). In contrast, a robust proliferation was observed in the

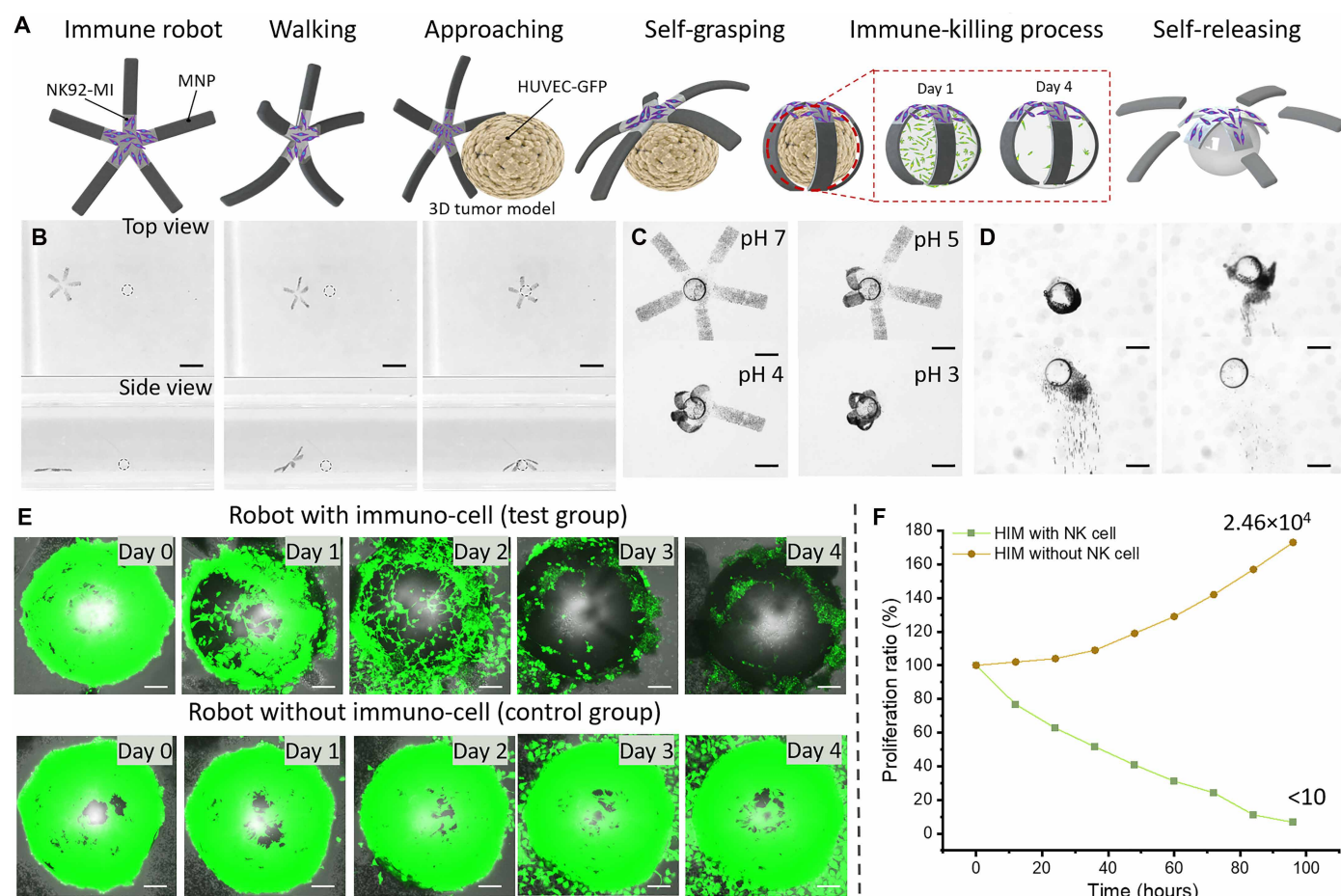


Fig. 4. In vitro immune-killing process of HIM with 3D tumor model. (A) Diagram of a comprehensive depiction of HIM's successive actions: advancing toward the 3D tumor model, enveloping it, seizing the tumor in response to pH changes, executing tumor destruction on the spheric structure, and subsequently dissolving under ionic stimulation. (B) Optical images capture HIM's locomotion toward and coverage of the 3D tumor model. Scale bars, 5 mm. (C) Optical images offer HIM's autonomous grasping of the 3D tumor model, triggered by pH changes within the microenvironment. Scale bars, 1 mm. (D) The sequence of optical images demonstrates HIM's precise removal of the 3D tumor postcompletion of the immune-killing process. Scale bars, 1 mm. (E) Fluorescent optical microscopy images provide a microscopic view of the immune-killing process enacted by NK92-MI cells after HIM grasps the 3D tumor model consisting of GFP-enhanced HUVECs. Scale bars, 300 μ m. (F) The cell survival ratio over time for the 3D tumor model treated with HIM, comparing outcomes with and without encapsulated immune cells.

group treated with empty HIMs (Fig. 4E). This showed that contact with the HIM did not obstruct the survival and proliferation of natural killer (NK) cells or HUVECs and provided evidence that the HIM was not only noncytotoxic but also did not interfere with the killing efficacy of immune cells. This, along with its excellent biocompatibility, underscored its medical potential as a cell carrier.

Ex vivo and in vivo locomotion

To evaluate the locomotion capabilities of the HIM in complex GI environments, we conducted a series of observations using ultrasound in an ex vivo setting, specifically within a rat intestine, as illustrated in Fig. 5A and demonstrated in movie S16. For this experiment, we embedded NK cells and MNPs into distinct modules of the HIM. The MNPs were placed in the leg modules to facilitate propulsion, enabling the millirobots to navigate through the intricate structures of the GI tract. As the HIM moved, it displayed an undulating locomotion pattern, which is well suited for maneuvering through the diverse and often challenging terrain of biological

tissues. The ultrasound imaging provided real-time insights into the movement dynamics of the HIM, capturing its initial positioning at the left end of the intestine at 25 s. Furthermore, to evaluate the in vivo locomotion performance of the HIM, we used digital subtraction angiography (DSA) to visualize its movement within the esophagus of a live rabbit. As shown in Fig. 5B and movie S17, the HIM was magnetically actuated under a DSA system, and its motion was monitored in real time. Time-lapse images captured from 0 to 25 s illustrate the trajectory of the HIM, marked by dashed circles. Despite the moderate imaging contrast caused by overlapping soft tissues, the HIM remained distinguishable to the naked eye throughout the experiment. These results confirmed the feasibility of controlled microrobot locomotion in vivo using clinical-grade imaging techniques. These observations not only highlighted the effective design and functionality of the HIM but also demonstrated its potential applications in targeted drug delivery and therapeutic interventions within the GI system. Through this study, we have demonstrated the feasibility of using the HIM in unstructured biological environments.

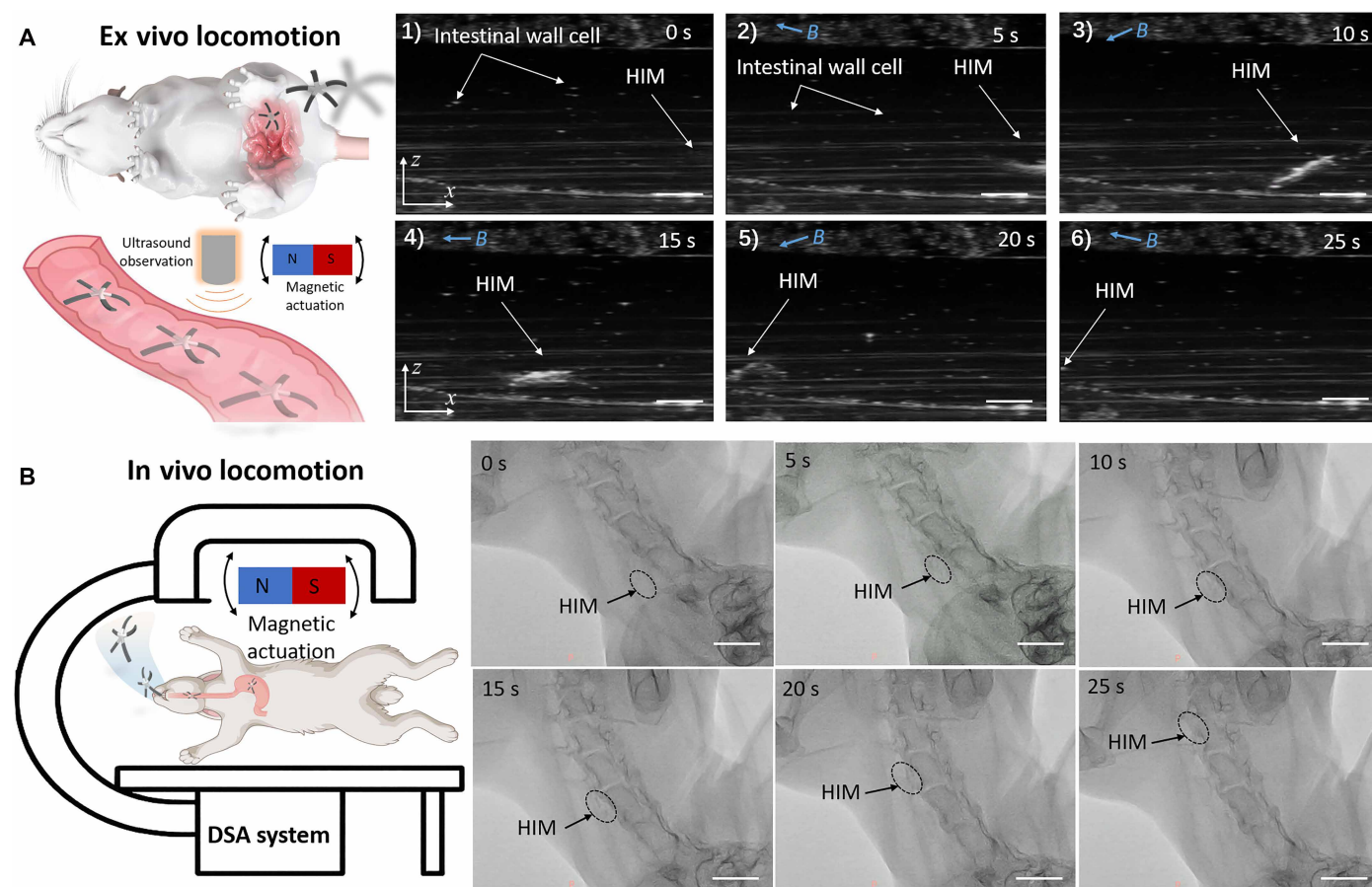


Fig. 5. Ex vivo and in vivo magnetic locomotion of the HIM inside the GI tract. (A) Ultrasound images for tracking the HIM locomotion under magnetic control. Scale bars, 2 mm. Ex vivo propulsion and target delivery of HIM carrying MNP and NK cells and controlled by external magnetic field. The HIM was manipulated using a magnetic system and observed using an ultrasound imaging system. (B) Schematic of the DSA imaging setup showing the magnetic actuation system used to drive the HIM inside the esophagus of a live rabbit. Time-lapse DSA images captured at 0, 5, 10, 15, 20, and 25 s after HIM insertion, showing its progression inside the esophagus. HIM was indicated by the dashed circles. Scale bars, 1 cm.

In vivo test of immune response

To explore HIM's applicability in digestive tract tumors in vivo, we focused on liver cancer and transplanted HIM with NK92-MI cells into HepG2-luc tumor-bearing mice. These mice were developed from BALB/c nude mice via subcutaneous injection of HepG2 cells, a human hepatoblastoma cell line. This model reflected the high incidence of hepatocellular carcinoma among individuals with chronic liver diseases like hepatitis B or hepatitis C infections. The progression from chronic hepatitis to liver cancer was closely tied to defects in NK cell function, which contributed to disease advancement and immune evasion by tumor cells. Individuals with hepatocellular carcinoma often showed decreased NK cell counts within tumors, as well as impaired recognition and killing capabilities (46, 47). Autologous NK cell therapy had shown promise in enhancing both the activity and quantity in peripheral circulation during the treatment of solid tumors (48). However, their limited migration to tumor sites posed challenges and hindered clinical efficacy.

We used HIM for NK cell transportation to test the potential in enhancing their ability to kill liver cancer cells by aiding in their physical recruitment within the body. The HEPG2-luc tumor-bearing mice were randomly assigned to three groups ($n = 5$ for

each group) to preliminarily assess the biocompatibility and the in vivo treatment effect of NK cells carried by the HIM. Mice that underwent surgery without further intervention were designated as the control group. Those transplanted with empty HIM were named the vehicle group, while mice receiving HIM carrying NK92-MI cells were identified as the test group. All mice that received HIM transplantation with or without NK cells lived well throughout the 15-day recording period. The tumor presence in each mouse was monitored via bioluminescence imaging using an In Vivo Imaging System (IVIS) every 4 days (Fig. 6A and fig. S13). The tumor in one mouse from the test group disappeared 8 days after the NK92-MI-HIM was transplanted, indicating the potential efficacy of HIM-carried NK cells. Photons emitted from the tumors were quantified as photons per second within regions of interest (Fig. 6B) and normalized to the values at day 0 (Fig. 6C). Tumor growth was rapid in the control group; by 12 days postsurgery, the tumors were notably larger than those in both the vehicle and test groups (Fig. 6C). This suggested a measurable immune response in both the vehicle and test groups.

To address this, we performed hematoxylin and eosin staining on tissues from wild-type mice at 6 hours, 24 hours, 3 days, 7 days,

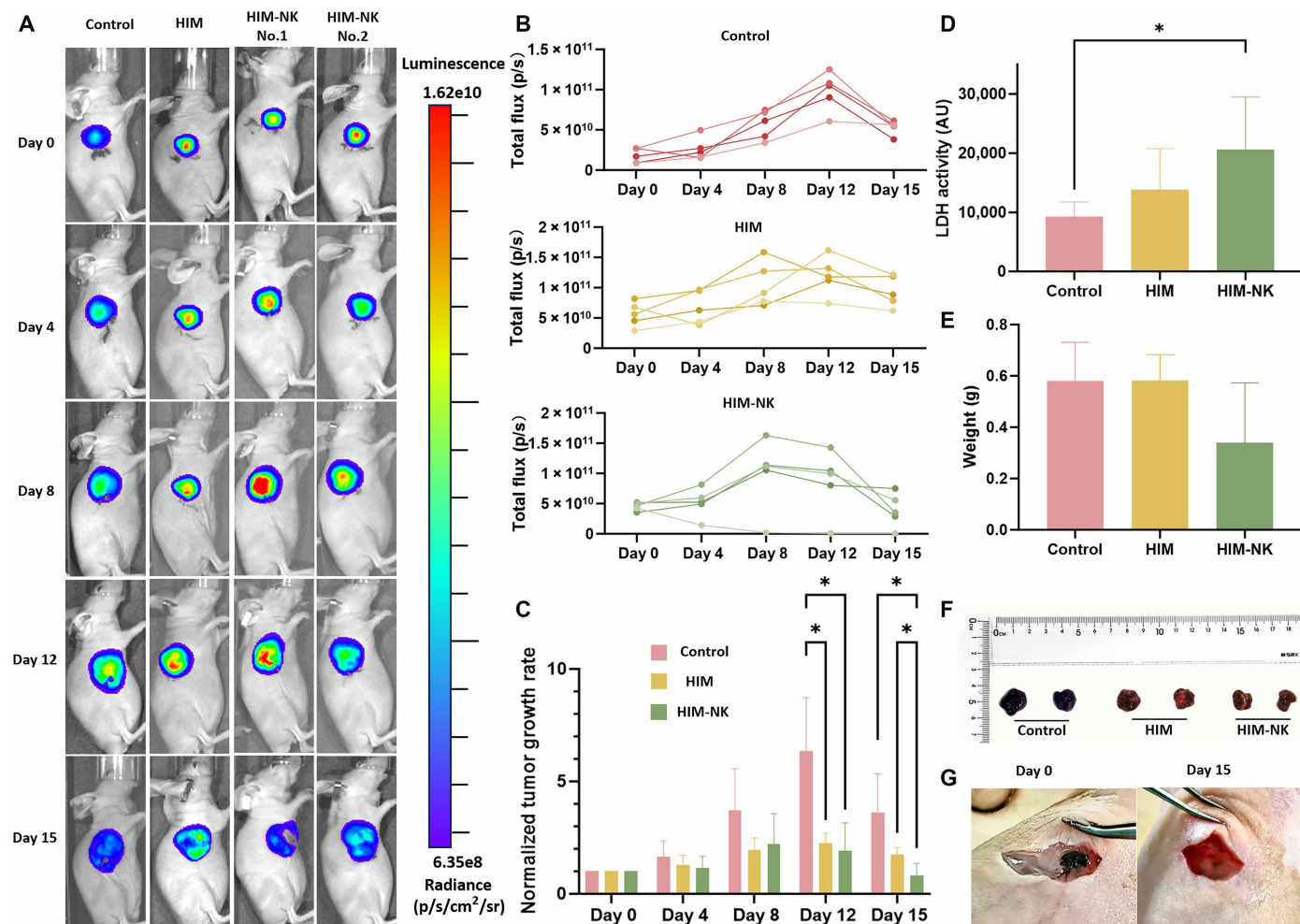


Fig. 6. Therapeutic efficacy of NK92-cell loaded HIM in HepG2-luc tumor-bearing mouse model. (A) Bioluminescence imaging of mice with subcutaneous tumors HEPG2-luc, with only surgery (Control), or treated by HIM carrying NK92-MI (HIM-NK) or not (HIM). The day when HIM was transplanted was set as day 0. (B) Luminescent intensity of photons emitted from each tumor was quantified, with each line representing an individual mouse. p/s, photons per second. (C) Luminescent intensity normalized by the value of day 0, statistical analysis showed significant differences between control with vehicle (* $P = 0.038$) and control with test group (* $P = 0.024$) at day 12 and between control with test (* $P = 0.044$) and vehicle with test (* $P = 0.034$) group at day 15. (D) Lactate dehydrogenase (LDH) activity assay in tumor tissues $\pm$ SEM ($n = 5$ for each group, $P = 0.026$). AU, absorbance unit. (E and F) The photographs (F) and weights (E; $n = 5$ for each group) of HepG2 tumors. (G) In the photographs of a representative in vivo subcutaneous tumor, the HIM structure is clear and distinct when it was first transplanted as shown on day 0 but degraded after 15 days.

and 14 days postimplantation with or without HIM transplantation (fig. S14). A transient inflammatory response was observed in the HIM group at the early time points. Specifically, increased inflammatory cell infiltration was noted at 6 and 24 hours postimplantation, relative to the surgery control group. However, by day 3, the inflammatory response in the HIM group had markedly subsided to a level comparable to the control group. By day 7, no significant difference was observed between the two groups. This indicated that, while the implantation of the HIM robot induced a short-term inflammatory response, such responses were self-limiting and resolved within 3 to 7 days without causing long-term damage or chronic inflammation. Together with the result from Fig. 6C, we hypothesized that, in the vehicle group, the transplantation of the HIM structure itself induced a mild immune response at the implantation site, recruiting the body's own repair mechanisms, which indirectly contributed to the killing of some tumor cells and inhibited

tumor growth. In comparison, the even lower tumor growth rate in the NK cell-loaded test group implied a stronger immune-killing effect facilitated by the NK cell-carrying HIM. To further evaluate potential tissue damage and biocompatibility, we measured lactate dehydrogenase (LDH) release in tissues collected 15 days postoperation. The LDH level in the vehicle group showed no significant difference compared with the control group (Fig. 6D), suggesting that the implanted HIM robot did not cause prolonged tissue damage and supporting its biocompatibility and safety in vivo. In contrast, a significantly higher LDH activity was observed in the test group compared to the control group (Fig. 6D), implying a stronger, targeted immune response. Tumors were collected and weighed on day 15 (Fig. 6, E and F). No traces of the HIM were found at the implantation site at this terminal time point (Fig. 6G), consistent with its biodegradability. These results collectively indicated the HIM's potential for in vivo use.

DISCUSSION

An increasing number of studies have shown that hydrogel soft robots held broad application prospects in the biomedical field. However, many problems remained to be solved in the process of transforming from the laboratory to clinical practice. First, the complex biological environment in the body required targeted medical robots to have movement and regulation modes that could adapt to different physiological scenarios such as pH. When pH was lower than 3, the HIM transformed into a spherical shape and conducted a rolling motion as illustrated in fig. S15 and movie S18, which confirmed its environmental adaptivity. The development of the HIM offered a promising strategy for enhancing immune response in a tumor study. Its modular design and ultrasoft structure endowed it with biomimetic, multimodal locomotion and adaptive shape morphing, which allowed it to navigate complex environments with ease. This versatility enabled it to access hard-to-reach tumor sites more effectively, thereby overcoming the limitations of conventional delivery methods. Unlike traditional cell delivery, which required cells to be manually spread across the target site, the HIM only needed to reach the desired location and maintain its position through self-adaptive morphing. This capability ensured localized stimulation and provided sufficient time for immune cells to secrete immunologic substances essential for effective tumor treatment. Furthermore, the hydrogel structure created a semi-isolated microenvironment, which supported immune cell functionality and safeguarded them as they performed their therapeutic tasks.

In vitro cytotoxicity studies have established that NK cells mediate tumor cell killing at effector-to-target (E:T) ratios as low as 1:1 to 2:1 against susceptible malignancies (49–51), with optimal activity typically achieved between 5:1 and 10:1 a therapeutic threshold often elevated within the immunosuppressive tumor microenvironment (52), highlighting the critical importance of localized delivery strategies. Our biohybrid robotic system successfully achieves and maintains a high localized NK cell density of ~ 4000 cells/mm² precisely at the tumor interface. Based on histological assessments indicating hepatocellular carcinoma cellularity ranges between 2000 and 4000 cells/mm² (53), this delivery capability establishes an effective local E:T ratio of 1:1 to 2:1 within the targeted treatment zone directly corresponding to the efficacious range demonstrated in conventional 2D cytotoxicity assays and substantially exceeding endogenous NK cell infiltration levels typically observed in hepatocellular carcinoma cores (54, 55). As quantitatively demonstrated in fig. S16, our HIM overcame the traditional engineering compromise between miniaturization and functional payload capacity that has constrained previous microrobotic systems (4, 22, 25, 42, 56, 57). Unlike microscale carriers that necessitate complex swarm coordination or repeated administrations due to severely limited volumetric capacity (21, 22, 44), the millimeter-scale HIM platform delivers therapeutically relevant biological cargo loads particularly crucial for transporting sensitive, nonproliferative cellular payloads like immune cells while simultaneously preserving exceptional softness and multimodal locomotor capabilities (including walking, tumbling, and undulating gaits) essential for navigation through complex and confined biological environments. This unique integration of substantial payload capacity and advanced trafficability positions the HIM platform as a promising technology for precision immune modulation in anatomically challenging therapeutic sites, including mucosal surfaces and postresection cavities, where localized immune activation is most needed.

By demonstrating the HIM's low toxicity, high biocompatibility, acceptable immune rejection, and biodegradability, we showed that this multimodal soft robotic gel structure held potential for cell delivery therapies. For example, compared to the commonly used CAR T cell targets in hematologic tumors, the existing antigen expression positivity rates in solid tumors were not high enough to recruit sufficient immune cells to the tumor site. Even if CAR T cells did gather around the tumor, it was difficult for them to penetrate the blood-tumor barrier; CAR T cell therapy targeting solid tumors also could easily lead to off-target effects and inadvertently damage normal tissues. To address these challenges, our HIM was able to physically grasp and target tumors, facilitating the precise recruitment of immune effectors. Its ability to navigate within GI bio-microenvironments, combined with its self-shaping capabilities in response to chemical and pH changes, made it a promising tool for investigating small-scale soft-bodied locomotion and enhancing the effectiveness of targeted therapies. With its multimodal locomotion capabilities, the HIM could navigate flexibly through the GI tract, facilitating early treatment of primary tumors. Moreover, its scalability supported a wide range of biomedical applications, from centimeter to sub-millimeter scales, which enhanced the versatility of HIM-based treatment approaches. The ultimate goal of the HIM was to embed patients' own immune cells to achieve immunity management for the treatment of tumors and other diseases.

Despite the successful demonstration of adaptive locomotion in a dynamic biological environment, several challenges must be addressed for clinical translation. As evidenced by the in vivo experiments, physiological disturbances including breathing, swallowing, and peristalsis compromised trajectory stability, while the nonlinear anatomy of the GI tract complicated directional control. These factors now limit the robot's positioning accuracy in unstructured environments. Furthermore, the HIM's small size and hydrogel composition resulted in poor visibility under clinical imaging, highlighting the need for material optimization with contrast-enhancing agents and integration with real-time imaging systems such as DSA. To overcome these limitations, future work will focus on developing closed-loop control systems that actively compensate for physiological motions and enhance positional accuracy. Additionally, testing in more complex animal models will be essential to evaluate long-term stability, biodegradability, and anatomical adaptability, ultimately advancing the system toward reliable minimally invasive therapeutic delivery.

MATERIALS AND METHODS

Fabrication of HIMs

The fabrication of HIMs began with the creation of alginate millistructures. A hydrogel deposition solution was prepared by mixing (1% w/v) sodium alginate with varying concentrations of CaCO₃ particles (0.5% w/v). This solution was then combined with MNPs at concentrations of (2% v/v) and immune cells (NK92-MI) to create different HIM modules. The MNPs consisted of FePt nanoparticles. To fabricate the millistructures, 2 ml of the hydrogel deposition solution was placed on an anode plate, with a cathode plate positioned on top, as illustrated in fig. S1. A direct current voltage ranging from 4 to 5 V was then applied to the fluorine-doped tin oxide layers of both plates for a duration of 10 s. After this electrodeposition process, the anode plate, still containing the hydrogel millistructures, was placed on a vibrating sample magnetometer (EZ7, Microsense, USA) to magnetize the HIMs. The HIMs were magnetized in the

horizontal direction using an external magnetic field of 0.8 T, which aligned the FePt nanoparticles to maximize the magnetic moment and driving force. Subsequently, the anode plate was rinsed with Hepes buffer in a 10-cm-diameter petri dish for over 3 min, ensuring that the calcium alginate hydrogel millistructures fully detached from the anode plate. Once detached, the millistructures were transferred to a 6-cm-diameter petri dish containing Hepes buffer, where they underwent further transformation.

Magnetic control system

The magnetic control system (fig. S17) for the HIMs generated a uniform magnetic field with a maximum strength of 20 mT within a workspace measuring 10 cm by 10 cm by 5 cm. This field was modulated by adjusting the currents in the electromagnetic coils, controlled by six independent motor driver units (SyRen25) and an Arduino microcontroller operating at 1.2 kHz. The motion of the HIMs was tracked using two cameras (aCa2040-90uc, Basler). The first camera, operating at 120 frames per second (fps) with a resolution of 2040 by 1020 pixels, was positioned orthogonally to the *yz* plane of the workspace to capture side views of the robot. The second camera, operating at 60 fps with a resolution of 2040 by 1400 pixels, provided a top-down view of the workspace.

Finite element analysis

FEA was conducted using commercial software (COMSOL, version 5.6, COMSOL Inc.) to understand the transformation behavior of the millistructures. A nonlinear elastic material model was used to simulate the structures' responses to changes in environmental pH. The 3D current density distribution (*C*) was calculated using the equation $C = \sigma E$ where σ represented the electroconductivity and *E* was the electric field strength. The swelling ratio was related to the current density. A 3D model was established and meshed using a superfine element type, and numerical analysis was carried out. The transformation of the 3D model included nonlinear transformation analysis using Newton's method. Similarly, FEA-based computer-aided design was carried out to describe the morphological changes in other microstructures before aniso-deposition as illustrated in fig. S2.

In vitro tests

The GFP-enhanced HUVECs were obtained from the School of Life Science and Technology at Beijing Institute of Technology. These cells were cultured in Dulbecco's modified Eagle's medium containing phenol red, 10% fetal bovine serum, and 1% penicillin-streptomycin (Gibco, Carlsbad, CA, USA). A LIVE/DEAD assay was conducted using calcein acetoxymethyl ester and propidium iodide (Dojindo Molecular Technologies Inc., Kumamoto, Japan) for fluorescein staining. The pH of the culture medium was adjusted to 6.0 to 6.5 for 1 to 2 min to soften the HIMs, allowing for the sequential separation of the magnetic module. HIM locomotion was controlled by the external magnetic system as described above. Before in vitro testing, glass spheres were coated with a layer of poly-lysine and collagen (Gibco, Carlsbad, CA, USA). The cell proliferation process was monitored by scanning the entire structure under a confocal microscope in a living cell workstation (A1, Nikon, Japan).

Ex vivo tests

The intestines were obtained from Sprague-Dawley rats received from Beijing Institute of Technology, Beijing, China. The animals were housed under room temperature of $20^\circ \pm 1^\circ\text{C}$, relative humidity

of $50 \pm 10\%$, and 12-hour light/12-hour dark cycle and received food and water ad libitum. The rats were euthanized, the abdominal cavity was opened, and the small and large intestines were removed from the stomach. The HIM was injected into the intestine that was filled with artificial intestinal liquid (Solarbio, Beijing, China), and the ends of the intestine were then sealed using two plastic clips. The HIM was manipulated by using a magnetic system. The manipulation process was observed using a Vevo 3100 LT imaging system (Fujifilm VisualSonics Inc., Canada).

In vivo locomotion

To evaluate the in vivo locomotion performance of the HIM, we developed a magnetic control guidance system incorporating a six-degree-of-freedom robotic arm (CR7-C, ROLIN Intelligent, China). This robotic arm offered a maximum end-load capacity of 7 kg and a high repetitive positioning accuracy of ± 0.02 mm, enabling stable and precise spatial control during dynamic actuation. A cubic N52-grade neodymium-iron-boron permanent magnet (50 mm by 50 mm by 100 mm) was securely mounted on the end effector of the robotic arm to generate a controllable 3D magnetic field. The magnetic field strength was measured to be 48.9 mT at a distance of 5 cm from the magnet surface, and the field gradient was quantified as 4.3 T/m at a distance of 2 to 5 cm. As shown in fig. S18, the HIM was actuated using an undulating magnetic field with a driving signal frequency of 0.2 to 1 Hz, sinusoidal or square waveform, and specified amplitude, with the magnet positioned at the side top of the robot to induce forward locomotion within the esophagus of a live rabbit. A healthy male New Zealand White rabbit (2.5 to 3.0 kg) was used for the in vivo imaging experiments. The animal was fasted for 12 to 16 hours before the procedure, with free access to water. Anesthesia was induced by intramuscular injection of a 1:1 mixture of BaiTe SuMianXin (0.1 g/ml) and Telazol (125 mg of tiletamine and 125 mg of zolazepam), administered according to body weight. Following full sedation, the HIM was introduced into the esophagus through a flexible feeding tube and guided by the external magnetic field generated by the robotic system. Real-time imaging of micro-robot motion was performed using a DSA system (Azurion 3 M15, Philips). After completion of the experiment, the animal was euthanized by intravenous overdose of anesthetic agents, in accordance with institutional animal care guidelines. Ethics approval for this study was granted under protocol number BIT-EC-SCXK (Beijing) 2021-0013-R-131.

In vivo immune response test

Female Balb/c nude mice (6 to 8 weeks old) received a subcutaneous injection of 5×10^6 HepG2-Luc cells on their back. Eight days post-inoculation, surgical transplantation of HIM was performed, designated as day 0. Tumor progression was monitored via fluorescence imaging starting 1-day postsurgery and subsequently every 3 days. For bioluminescence imaging, D-luciferin (15 mg/ml) was dissolved in Dulbecco's Phosphate Buffered Saline, sterilized by 0.2- μm filtration, and administered intraperitoneally at 10 $\mu\text{l/g}$ of body weight. Imaging was conducted 15 min postinjection using the IVIS Spectrum system. Fifteen days after surgery, the mice were euthanized by cervical dislocation, and the tumors were excised for weighing, photography, and LDH analysis.

All surgical and experimental procedures were approved by the Institutional Animal Care and Use Committee of the Beijing Institute of Technology, Beijing, China. Animals were housed under standardized

and sterile conditions (room temperature of $20^{\circ} \pm 1^{\circ}\text{C}$, relative humidity of $50 \pm 10\%$, and 12/12 hours light/dark cycle) and received food and water ad libitum.

Supplementary Materials

The PDF file includes:

Supplementary Text

Figs. S1 to S18

Legends for movies S1 to S18

Other Supplementary Material for this manuscript includes the following:

Movies S1 to S18

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